MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

Antibiotic Treatment Guidelines for Acute Otitis Media in Pediatric Patients

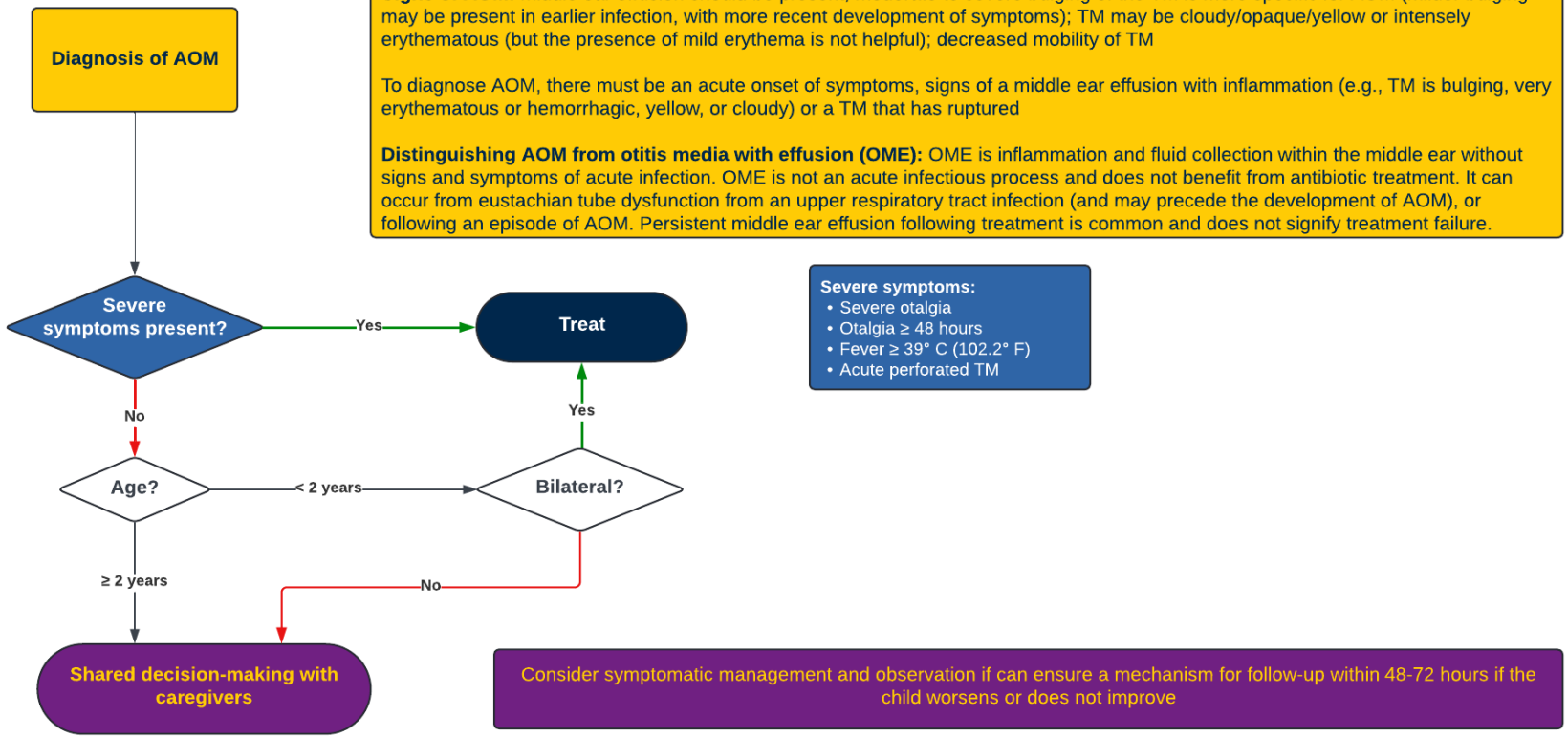
This guideline is designed to provide guidance for uncomplicated acute otitis media (AOM) in most children. The guideline assumes that the clinician has made an accurate diagnosis of AOM. Management of AOM in patients < 3 months of age or in patients with certain comorbidities (e.g., tympanostomy tubes or chronically perforated tympanic membrane, cochlear implant) or with presence of suppurative complications (such as mastoiditis) is beyond the scope of these guidelines. Management of chronic otitis media or chronic otorrhea is also beyond the scope of these guidelines.

Symptoms of AOM: fever, ear pain (in a preverbal child, this may present as ear tugging/rubbing/holding and/or excessive crying), irritability, difficulty sleeping, decreased activity or appetite; symptoms are nonspecific – symptoms alone without signs of AOM are insufficient to diagnose AOM

Signs of AOM: middle ear effusion should be present; moderate to severe bulging of the TM is more specific for AOM (milder bulging may be present in earlier infection, with more recent development of symptoms); TM may be cloudy/opaque/yellow or intensely erythematous (but the presence of mild erythema is not helpful); decreased mobility of TM

To diagnose AOM, there must be an acute onset of symptoms, signs of a middle ear effusion with inflammation (e.g., TM is bulging, very erythematous or hemorrhagic, yellow, or cloudy) or a TM that has ruptured

Distinguishing AOM from otitis media with effusion (OME): OME is inflammation and fluid collection within the middle ear without signs and symptoms of acute infection. OME is not an acute infectious process and does not benefit from antibiotic treatment. It can occur from eustachian tube dysfunction from an upper respiratory tract infection (and may precede the development of AOM), or following an episode of AOM. Persistent middle ear effusion following treatment is common and does not signify treatment failure.



Observation for initial management of AOM should be considered whenever possible; observation does not increase suppurative complications if follow-up is provided and delayed antibiotics are begun for persistent or worsening symptoms. Children < 2 years with AOM may take longer to improve clinically than older children.

When to refer to ED: Children with evidence of suppurative complications (such as mastoiditis) or toxic appearance

When to refer to ENT: Recurrent otitis media (3 or more well-documented and separate AOM episodes in the preceding 6 months *or* 4 or more episodes in the preceding 12 months with at least 1 episode in the past 6 months), chronic otitis media/persistent otitis media with effusion (> 3 months)

When to refer to ID: Recurrent/chronic episodes of otitis media unresponsive to standard therapy, unusual or antibiotic-resistant organisms recovered from middle ear fluid (if advice needed acutely, page on-call physician), concern for immunodeficiency as a cause of recurrent infections

Indication	Empiric Therapy	Duration/Comments
Initial treatment Common pathogens: *S. pneumoniae* *H. influenzae* *M. catarrhalis*	1st line: **Amoxicillin*** 45 mg/kg/DOSE PO q12h (max: 2 g/dose) Indications for amoxicillin-clavulanate - Received amoxicillin or amoxicillin-clavulanate in the past 30 days - Concurrent purulent conjunctivitis ("otitis-conjunctivitis") **Amoxicillin-clavulanate*** 45 mg/kg/DOSE PO q12h (max: 2 g/dose) Received oral cephalosporin in past 30 days: **Amoxicillin*** 45 mg/kg/DOSE PO q12h (max: 2 g/dose) – see comment Penicillin allergy:[1] *Preferred:* **Ceftriaxone** 50 mg/kg/DOSE IM once (max: 1 g/dose) *Alternative* (oral cephalosporins have lower activity against *S. pneumoniae* than ceftriaxone; consider if ceftriaxone is not logistically possible): **Cefuroxime*** 15 mg/kg/DOSE q12h (max: 500 mg/dose) OR **Cefdinir*** 7 mg/kg/DOSE q12h (max: 300 mg/dose) High-risk cephalosporin allergy, non-IgE mediated allergy, or other inability to take oral cephalosporins:[2,3] **Levofloxacin***: < 5 years: 10 mg/kg/DOSE IV/PO BID (max: 375 mg/DOSE) ≥ 5 years: 10 mg/kg/DOSE IV/PO daily (max: 750 mg/DOSE)	• Duration for oral medications: ○ **< 2 years: 10 days** ○ **≥ 2 years: 5 days** • If there is a history of AOM not responsive to amoxicillin, it is reasonable to try amoxicillin first, if it has been >30 days since the most recent infection • High-dose amoxicillin is preferred over oral cephalosporins, especially cefdinir, for otitis media caused by *S. pneumoniae* due to superior PK/PD parameters • Michigan Medicine 2022 susceptibility rates for S. pneumoniae: penicillin 97%, cefuroxime 86%, ceftriaxone 97% • Cefuroxime suspension is no longer available in the US; tablets may be crushed • Use amoxicillin-clavulanate ES (600 mg/42.9 mg/5 mL) suspension to limit the risk of diarrhea associated with high doses of clavulanate • In case of penicillin allergy, review β-lactam allergy evaluation and empiric guidance and remove allergy label if possible. If not, consider referral to Allergy clinic to provide guidance on future episodes.
Treatment failure (Patients who were compliant with and tolerated appropriate initial treatment for > 48 hours)	Failed amoxicillin: **Amoxicillin-clavulanate*** 45 mg/kg/DOSE PO q12h (max: 2 g/dose) Failed amoxicillin-clavulanate or oral cephalosporin: **Ceftriaxone** 50 mg/kg/DOSE IM daily for 3 days (max: 1 g/dose) OR **Levofloxacin***: < 5 years: 10 mg/kg/DOSE IV/PO BID (max: 375 mg/DOSE) ≥ 5 years: 10 mg/kg/DOSE IV/PO daily (max: 750 mg/DOSE) Failed all above options: Consult Pediatric Infectious Diseases	• Duration for oral medications: ○ **< 2 years: 10 days** ○ **≥ 2 years: 5 days** • 3 days of ceftriaxone has been shown to be superior to 1 day for penicillin resistant *S. pneumoniae* • Decision between ceftriaxone and levofloxacin can be based on logistics and provider/parent preference; levofloxacin suspension may require prior authorization and/or not be readily available at community pharmacies; levofloxacin tablets may be crushed

* Renal adjustment may be necessary. See Pediatric Antimicrobial Dosing Guidelines.

[1] Low-risk allergies include: pruritus without rash, remote (> 10 years) unknown reaction, mild to severe rash with no other symptoms; see β-lactam allergy evaluation and empiric guidance for further information

[2] High-risk allergies include: anaphylaxis, respiratory symptoms (chest tightness, bronchospasm, wheezing, cough), angioedema (swelling, throat tightness), cardiovascular symptoms (hypotension, dizzy/lightheadedness, syncope/passing out, arrhythmia); see β-lactam allergy evaluation and empiric guidance for further information.

[3] Non-IgE mediated allergies are contraindications to further beta-lactam use unless approved by Allergy: organ damage (kidney, liver), drug-induced immune-mediated anemia/thrombocytopenia/leukopenia, rash with mucosal lesions (Stevens Johnson Syndrome/Toxic Epidermal Necrosis), rash with pustules (acute generalized exanthematous pustulosis), rash with eosinophils and organ injury (DRESS – drug rash eosinophilia and systemic symptoms), rash with joint pain, fever, and myalgia (Serum Sickness); see β-lactam allergy evaluation and empiric guidance for further information

Authors:

Elizabeth Lloyd (Pediatric Infectious Diseases / Antimicrobial Stewardship)
Karen Davidge (Pharmacy / Antimicrobial Stewardship)
Nicholas Dillman (Pharmacy / Antimicrobial Stewardship)
Margeaux Naughton (General Pediatrics)
Julia Madison-Williams (General Pediatrics)
Elizabeth Jahns (Internal Medicine-Pediatrics)
Leigh Morrison (Family Medicine)
Natalie Schellpfeffer (Pediatric Emergency Medicine)

Consultants:

Marc Thorne (Pediatric Otolaryngology)

Last revised: 10/2024	**Antimicrobial Stewardship Subcommittee:** 10/2024	**P&T:** 11/2024

References

	Citation
1	Lieberthal AS, Carroll AE, Chonmaitree T, Ganiats TG, Hoberman A, Jackson MA, Joffe MD, Miller DT, Rosenfeld RM, Sevilla XD, Schwartz RH, Thomas PA, Tunkel DE. The diagnosis and management of acute otitis media. Pediatrics. 2013 Mar;131(3):e964-99. doi: 10.1542/peds.2012-3488. Epub 2013 Feb 25. Erratum in: Pediatrics. 2014 Feb;133(2):346. Dosage error in article text. PMID: 23439909.
2	Le Saux N, Robinson JL; Canadian Paediatric Society, Infectious Diseases and Immunization Committee. Management of acute otitis media in children six months of age and older. Paediatr Child Health. 2016 Jan-Feb;21(1):39-50. doi: 10.1093/pch/21.1.39. PMID: 26941560; PMCID: PMC4758427.
3	Venekamp RP, Sanders SL, Glasziou PP, Rovers MM. Antibiotics for acute otitis media in children. Cochrane Database Syst Rev. 2023 Nov 15;11(11):CD000219. doi: 10.1002/14651858.CD000219.pub5. PMID: 37965923; PMCID: PMC10646935.
4	Kozyrskyj A, Klassen TP, Moffatt M, Harvey K. Short-course antibiotics for acute otitis media. Cochrane Database Syst Rev. 2010 Sep 8;2010(9):CD001095. doi: 10.1002/14651858.CD001095.pub2. PMID: 20824827; PMCID: PMC7052812.
5	Hoberman A, Paradise JL, Rockette HE, Kearney DH, Bhatnagar S, Shope TR, Martin JM, Kurs-Lasky M, Copelli SJ, Colborn DK, Block SL, Labella JJ, Lynch TG, Cohen NL, Haralam M, Pope MA, Nagg JP, Green MD, Shaikh N. Shortened Antimicrobial Treatment for Acute Otitis Media in Young Children. N Engl J Med. 2016 Dec 22;375(25):2446-2456. doi: 10.1056/NEJMoa1606043. PMID: 28002709; PMCID: PMC5319589.

Revision History

Date	Revision

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.